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Command of Evidence

08

9 questions · ~14 min

Partial List of Candidate Species for De-extinction

Common name	Scientific name	Became extinct
Huia	<i>Heteralocha acutirostris</i>	1907
Caribbean monk seal	<i>Monachus tropicalis</i>	1952
Passenger pigeon	<i>Ectopistes migratorius</i>	1914
Saber-toothed cat	<i>Smilodon</i>	11,000 years before present
Woolly mammoth	<i>Mammuthus primigenius</i>	6,400 years before present

The passage of time is among the many obstacles faced by scientists who are pursuing de-extinction efforts—that is, efforts to use breeding or a mixture of cloning and genetic engineering to bring back extinct species. Specifically, researchers are concerned that the longer a species has been extinct, the less likely it is that a suitable habitat still exists for that species. Among candidate species for de-extinction, this problem would be especially concerning for the _____ blank

Which choice most effectively uses data from the table to complete the statement?

- A. passenger pigeon (*Ectopistes migratorius*), which became extinct only a few years after the huia (*Heteralocha acutirostris*).
- B. saber-toothed cat (*Smilodon*), which became extinct 11,000 years ago.

- C. woolly mammoth (*Mammuthus primigenius*), which became extinct several thousand years before the saber-toothed cat (*Smilodon*).
- D. Caribbean monk seal (*Monachus tropicalis*), which became extinct in 1952.

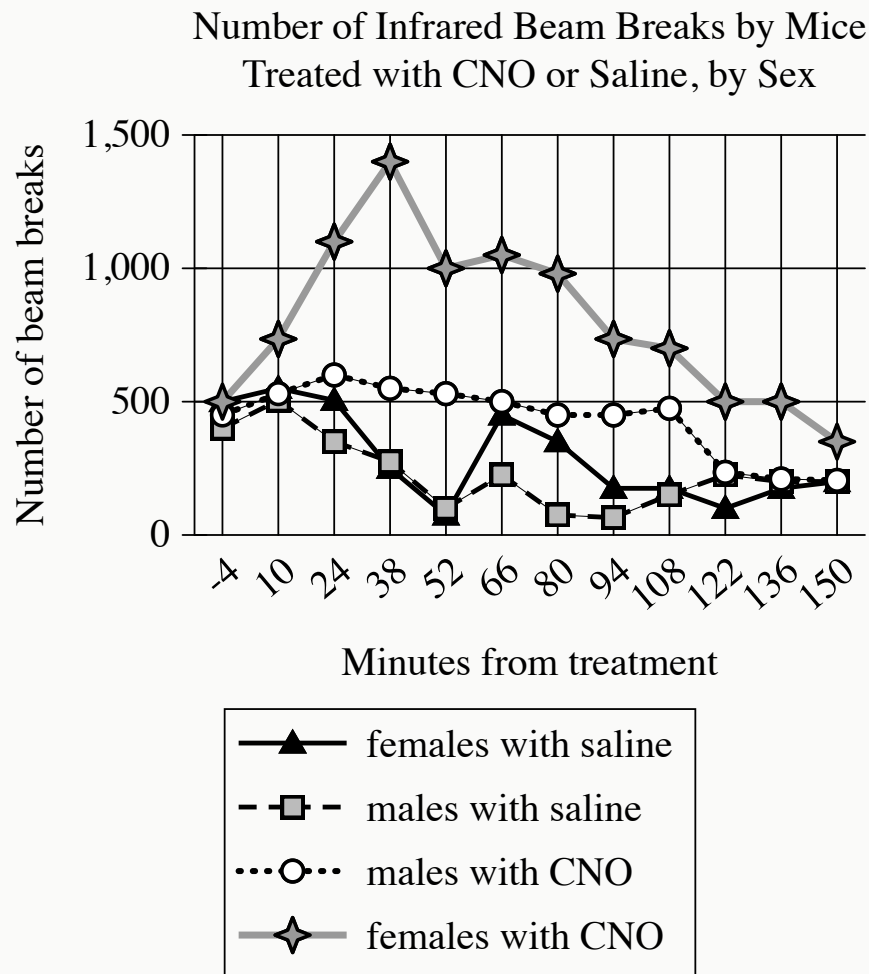
Average Nitrate and Phosphate Concentrations in Seawater after Volcanic Eruption

Nutrient	Seawater in lava-affected area, 5–45 meters below surface	Seawater in lava-affected area, 75–125 meters below surface	Seawater outside of lava-affected area, 5–45 meters below surface	Seawater outside of lava-affected area, 75–125 meters below surface
Nitrate (micromoles per liter)	3.1	0.4	≤0.03less than or equal to 0.03	≤0.01less than or equal to 0.01
Phosphate (micromoles per liter)	0.17	0.09	0.14	0.06

After a volcanic eruption spilled lava into North Pacific Ocean waters, a dramatic increase of diatoms (a kind of phytoplankton) near the surface occurred. Scientists assumed the diatoms were thriving on nutrients such as phosphate from the lava, but analysis showed these nutrients weren't present near the surface in forms diatoms can consume. However, there was an abundance of usable nitrate, a nutrient usually found in much deeper water and almost never found in lava. Microbial oceanographer Sonya Dyhrman and colleagues believe that as the lava plunged nearly 300 meters below the surface it dislodged pockets of this nutrient, releasing it to float upward, given that _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. at 5–45 meters below the surface, the average concentration of phosphate was about the same in the seawater in the lava-affected area as in the seawater outside of the lava-affected area.
- B. for both depth ranges measured, the average concentrations of nitrate were substantially higher in the seawater in the lava-affected area than in the seawater outside of the lava-affected area.
- C. for both depth ranges measured in the seawater in the lava-affected area, the average concentrations of nitrate were substantially higher than the average concentrations of phosphate.
- D. in the seawater outside of the lava-affected area, there was little change in the average concentration of nitrate from 75–125 meters below the surface to 5–45 meters below the surface.



- All values are approximate.
- The following 4 lines are shown:
 - females with CNO
 - females with saline
 - males with CNO
 - males with saline
- The females with CNO line:
 - Begins at negative 4, 500
 - Rises sharply to 10, 735
 - Rises sharply to 24, 1,100
 - Rises sharply to 38, 1,400

- Falls sharply to 52, 1,000
- Rises gradually to 66, 1,050
- Falls gradually to 80, 980
- Falls sharply to 94, 735
- Falls gradually to 108, 700
- Falls sharply to 122, 500
- Remains level to 136, 500
- Falls sharply to 150, 350
- The females with saline line:
 - Begins at negative 4, 500
 - Rises gradually to 10, 550
 - Falls gradually to 24, 505
 - Falls sharply to 38, 250
 - Falls sharply to 52, 75
 - Rises sharply to 66, 450
 - Falls gradually to 80, 350
 - Falls sharply to 94, 175
 - Remains level to 108, 175
 - Falls gradually to 122, 100
 - Rises gradually to 136, 175
 - Rises gradually to 150, 200
- The males with CNO line:
 - Begins at negative 4, 450
 - Rises gradually to 10, 530
 - Rises gradually to 24, 600
 - Falls gradually to 38, 550
 - Falls gradually to 52, 530
 - Falls gradually to 66, 500
 - Falls gradually to 80, 450
 - Remains level to 94, 450
 - Rises gradually to 108, 475
 - Falls sharply to 122, 235
 - Falls gradually to 136, 210
 - Falls gradually to 150, 205
- The males with saline line:
 - Begins at negative 4, 400
 - Rises sharply to 10, 505
 - Falls sharply to 24, 350

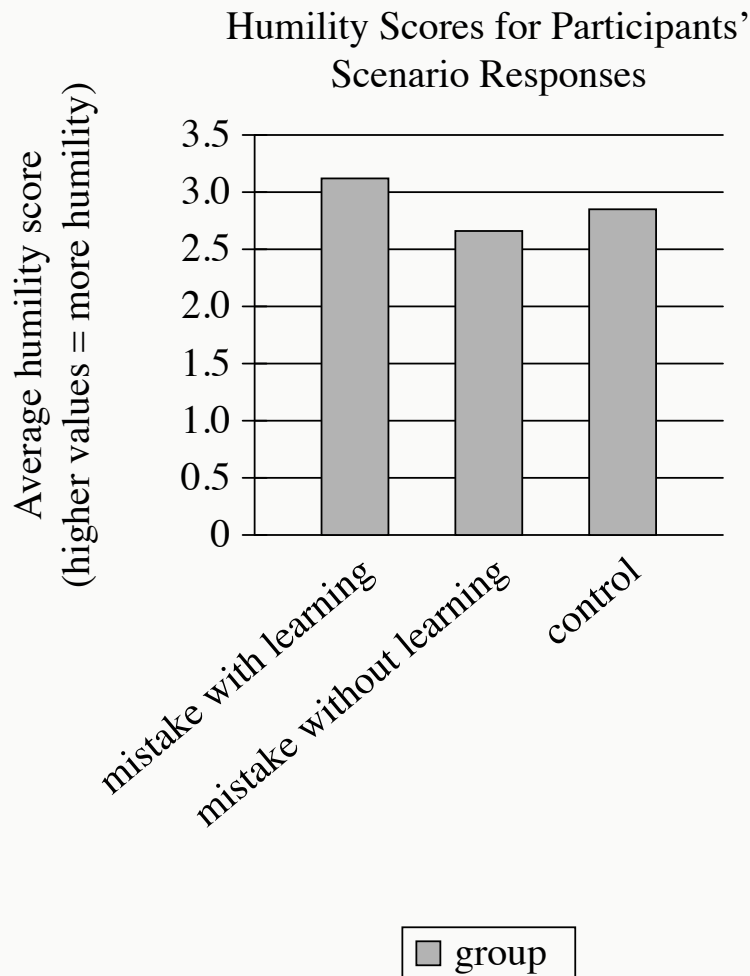
- Falls gradually to 38, 275
- Falls sharply to 52, 100
- Rises sharply to 66, 225
- Falls sharply to 80, 75

To investigate the influence of certain estrogen-responsive neurons on energy expenditure, biologist Stephanie Correa et al. treated female and male mice with either saline solution or clozapine-N4-oxide (CNO), which activates the neurons. Monitoring the activity levels of the mice by measuring how frequently the animals broke infrared beams crossing their enclosures, Correa et al. found that the mice in their study showed sex-specific differences in response to neuron activation:

_____blank

Which choice most effectively uses data from the graph to complete the assertion?

- A. the four groups of mice differed greatly in their activity levels before treatment but showed identical activity levels at the end of the monitoring period.
- B. saline-treated females showed substantially more activity at certain points in the monitoring period than saline-treated males did.
- C. CNO-treated females showed more activity relative to saline-treated females than CNO-treated males showed relative to saline-treated males.
- D. CNO-treated females showed a substantial increase and then decline in activity over the monitoring period, whereas CNO-treated males showed a substantial decline in activity followed by a steep increase.



- The Average humility score data for the 3 bars are as follows:
 - mistake with learning: 3.12
 - mistake without learning: 2.66
 - control: 2.85

Jia Hu and colleagues hypothesized that workplace leaders who reflect on lessons learned from past mistakes are likely to exhibit more humility than leaders who don't engage in such reflection. To test this, the team placed 301 managers in one of three groups. Participants in two experimental groups were asked to reflect on a mistake, one group focusing on a mistake that resulted in learning and the other group focusing on a mistake that didn't result in learning. Participants in a control

group were asked to reflect on their daily routine. All the participants then described how they would respond to a workplace scenario. After evaluating the responses for evidence of humility, the researchers concluded that their hypothesis was correct.

Which choice best describes data in the graph that support the researchers' conclusion?

- A. None of the three groups' average humility scores exceeded 3.5.
- B. The managers in the control group exhibited only slightly less humility on average than the managers in the two experimental groups did.
- C. The managers who reflected on a past mistake that resulted in learning exhibited more humility on average than the managers in the other two groups did.
- D. All three groups exhibited less humility on average than the researchers expected.

There Is Confusion is a 1924 novel by Jessie Redmon Fauset. In the novel, the narrator portrays the character Joanna as someone who admires ambition in other people to the exclusion of all other qualities: _____ blank

Which quotation from *There Is Confusion* most effectively illustrates the claim?

- A. “Joanna was mightily interested in people who had a ‘purpose’ in life. Otherwise not at all.”
- B. “Indeed from the very beginning Joanna showed her preference for her father.”
- C. “Joanna was like her father not only so far as ambition was concerned but also in her willingness to work.”
- D. “She had a good sense of logic, a strong power of concentration, and a remarkably retentive and visualizing memory.”

External shopping cues are a type of marketing that uses obvious messaging—a display featuring a new product, for example, or a “buy one, get one free” offer—to entice consumers to make spontaneous purchases. In a study, data scientist Sam K. Hui and colleagues found that this effect can also be achieved with a less obvious cue: rearranging a store’s layout. The researchers explain that trying to find items in new locations causes shoppers to move through more of the store, exposing them to more products and increasing the likelihood that they’ll buy an item they hadn’t planned on purchasing.

Which response from a survey given to shoppers who made a purchase at a retail store best supports the researchers’ explanation?

- A. “I needed to buy some cleaning supplies, but they weren’t in their regular place. While I was looking for them, I saw this interesting notebook and decided to buy it, too.”
- B. “I didn’t buy everything on my shopping list today. I couldn’t find a couple of the items in the store, even though I looked all over for them.”
- C. “The store sent me a coupon for a new brand of soup, so I came here to find out what kinds of soup that brand offers. I decided to buy a few cans because I had the coupon.”
- D. “This store is larger than one that’s closer to where I live, and it carries more products. I came here to buy some things that the other store doesn’t always have.”

North American Thrasher Mean Bill Size and Habitat Temperature
Range

Species	Mean bill surface area (cm ² cm squared)	Mean maximum temperature of warmest month (°C)	Mean minimum temperature of coldest month (°C)
Brown thrasher	1.86	30.40	-4.29
Bendire's thrasher	1.98	36.57	0.24
Long-billed thrasher	2.24	35.27	8.82
Cozumel thrasher	2.28	33.27	18.21
Ocellated thrasher	3.26	27.56	5.45

It has been hypothesized that since birds can dissipate excess heat through their bills, bill size should increase with habitat temperature. To evaluate this hypothesis for a 2021 study, Charlotte Probst and colleagues gathered data on mean bill surface area of species of North American thrashers (genus: *Toxostoma*) as well as on climate conditions of the birds' native habitats. Based on their data, Probst and colleagues concluded that the hypothesis was not fully supported.

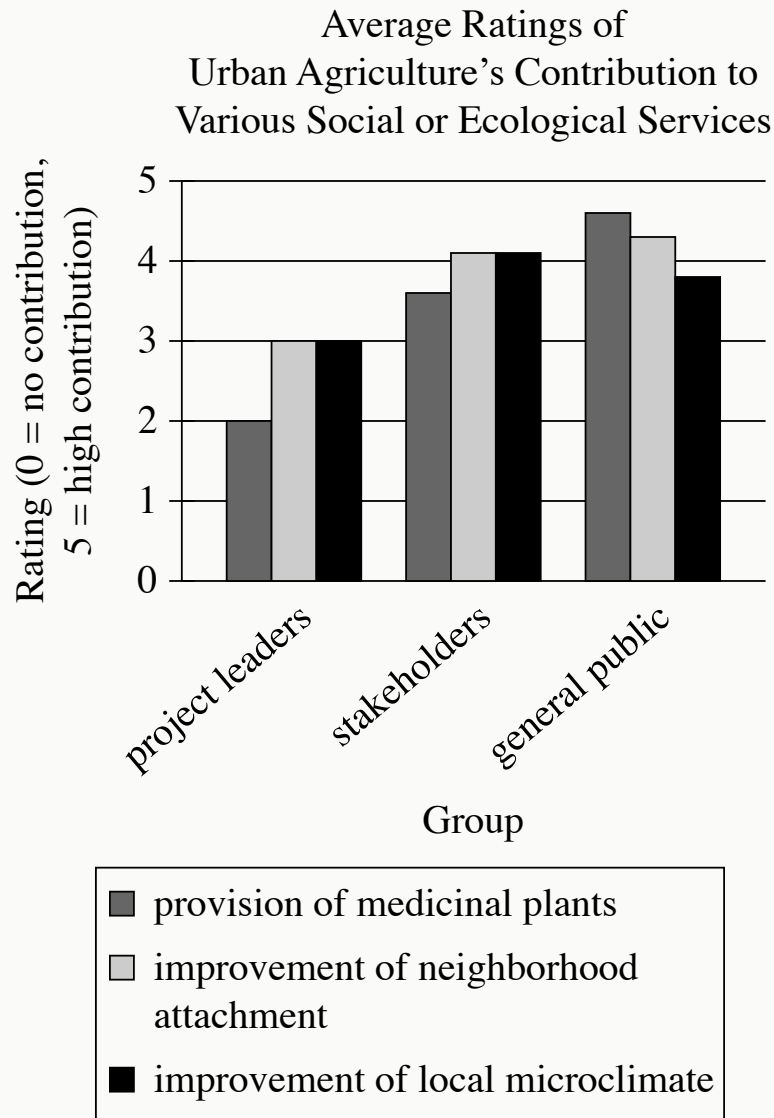
Which choice most effectively uses data from the table to support Probst and colleagues' conclusion?

- A. Although the Bendire's thrasher has one of the smallest mean bill surface areas of the birds included in the table, its habitat has one of the lowest mean maximum temperatures in the warmest month and one of the lowest mean minimum temperatures in the coldest month.
- B. Although the Cozumel thrasher has the second greatest mean bill surface area of the birds included in the table, its habitat's mean temperature in the warmest month is significantly higher than that of the other birds' habitats.
- C. Of the birds included in the table, the brown thrasher has the smallest mean bill surface area and the habitat with the lowest mean minimum temperature in the coldest month, while the long-billed thrasher has the second-largest mean bill surface area and the habitat with the second-highest mean minimum temperature in the coldest month.
- D. Of the birds included in the table, the ocellated thrasher has the largest mean bill surface area and the habitat with the lowest mean maximum temperature in the warmest month, while the Bendire's thrasher has the second smallest mean bill surface area and the habitat with the highest mean maximum temperature in the warmest month.

Hedda Gabler is an 1890 play by Henrik Ibsen. As a woman in the Victorian era, Hedda, the play's central character, is unable to freely determine her own future. Instead, she seeks to influence another person's fate, as is evident when she says to another character, _____ blank

Which quotation from a translation of *Hedda Gabler* most effectively illustrates the claim?

- A. "Then what in heaven's name would you have me do with myself?"
- B. "I want for once in my life to have power to mould a human destiny."
- C. "Then I, poor creature, have no sort of power over you?"
- D. "Faithful to your principles, now and for ever! Ah, that is how a man should be!"



- For each data category, the following bars are shown:
 - provision of medicinal plants
 - improvement of neighborhood attachment
 - improvement of local microclimate
- The data for the 3 categories are as follows:
 - project leaders:
 - provision of medicinal plants: 2.0
 - improvement of neighborhood attachment: 3.0

- improvement of local microclimate: 3.0
- stakeholders:
 - provision of medicinal plants: 3.6
 - improvement of neighborhood attachment: 4.1
 - improvement of local microclimate: 4.1
- general public:
 - provision of medicinal plants: 4.6
 - improvement of neighborhood attachment: 4.3
 - improvement of local microclimate: 3.8

Urban agriculture is the practice of growing plant- or animal-based products in urban settings such as community gardens and rooftop farms. Esther Sanyé-Mengual, Kathrin Specht, and their team surveyed three groups of people in Bologna, Italy—leaders of urban agriculture projects, stakeholders in urban agriculture (e.g., food researchers and urban farming associations), and the general public—to compare their views about the extent to which urban agriculture contributes to 25 social or ecological services that the team identified. The survey results show that, on average, project leaders rated urban agriculture as contributing less to _____blank

Which choice most effectively uses data from the graph to complete the assertion?

- A. providing medicinal plants than to improving the local microclimate.
- B. improving community members' sense of attachment to the neighborhood than to offering educational opportunities.
- C. improving the local microclimate than to providing medicinal plants.
- D. improving the local microclimate than to providing tourist attractions.